

Quantitative Aptitude Set-05

Directions (Q1–Q6): Study the following data carefully and answer the questions.

The table shows the total population (males + females) of four cities A, B, C and D in 2024 and 2025.

City	2024	2025
A	360	630
B	500	760
C	610	690
D	924	450

Q1. The ratio of males to females in city A in 2024 is 2:3. The number of females in city A in 2025 is 1.5 times the number of females in 2024. Find the difference between the number of males in city A in 2024 and 2025.

- (a) 172
- (b) 162
- (c) 148
- (d) 186
- (e) 156

Answer: (b) 162

Solution:

Males in A in 2024 = $360 \times \frac{2}{5} = 144$
Females in 2024 = $360 - 144 = 216$
Females in 2025 = $216 \times 1.5 = 324$
Males in 2025 = $630 - 324 = 306$
Required difference = $306 - 144 = 162$

Q2. The population of city Z in 2024 is 30% more than that of city B in 2024. If the total population of city Z in 2024 and 2025 together is 1250, find its population in 2025.

- (a) 580
- (b) 620
- (c) 600
- (d) 650
- (e) 560

Answer: (c) 600

Solution:

Population of Z in 2024 = $500 \times \frac{130}{100} = 650$
Population of Z in 2025 = $1250 - 650 = 600$

Q3. The population of city B in 2023 was 20% less than the average population of cities A and C in 2025. Find the ratio of the population of B in 2023 to that of D in 2024.

- (a) 3:7

- (b) 5:9
- (c) 4:7
- (d) 5:8
- (e) 6:11

Answer: (c) 4:7

Solution:

Average population of A and C in 2025 = $(630 + 690)/2 = 660$

B's population in 2023 = 80% of 660 = 528

Required ratio = $528 : 924 = 4 : 7$

Q4. The number of males in city D in 2024 is 50% of the total population of city B in 2025. The number of males in D in 2025 is half the number of males in D in 2024. Find the total number of females in D in 2024 and 2025 together.

- (a) 784
- (b) 814
- (c) 794
- (d) 804
- (e) 774

Answer: (d) 804

Solution:

Males in D in 2024 = 50% of 760 = 380

Females in D in 2024 = $924 - 380 = 544$

Males in D in 2025 = $380/2 = 190$

Females in D in 2025 = $450 - 190 = 260$

Required total = $544 + 260 = 804$

Q5. The population of city C in 2021 was 75% of the population of city A in 2024. If 40% of the population of C in 2021 were males, find the number of females as a percentage of the population of city D in 2025.

- (a) 32%
- (b) 40%
- (c) 36%
- (d) 30%
- (e) 44%

Answer: (c) 36%

Solution:

Population of C in 2021 = 75% of 360 = 270

Males = 40% of 270 = 108

Females = $270 - 108 = 162$

Required percentage = $162/450 \times 100 = 36\%$

Q6. $\frac{3}{4}$ of the population of city D in 2024 are literate, while $\frac{7}{20}$ of the population of city B in 2024 are illiterate. Find the difference between the total literate population of B and D together and the total illiterate population of B and D together.

- (a) 592
- (b) 602
- (c) 622
- (d) 632
- (e) 612

Answer: (e) 612

Solution:

$$\text{Literate in D} = 924 \times \frac{3}{4} = 693$$

$$\text{Illiterate in D} = 924 - 693 = 231$$

$$\text{Illiterate in B} = 500 \times \frac{7}{20} = 175$$

$$\text{Literate in B} = 500 - 175 = 325$$

$$\text{Total literates} = 693 + 325 = 1018$$

$$\text{Total illiterates} = 231 + 175 = 406$$

$$\text{Difference} = 1018 - 406 = 612$$

Directions (Q7–Q10): Read the following information carefully.

There are 249 students in three colleges P, Q and R.

Number of girls in Q = 40.

Boys in P are 55% of girls in Q.

Girls in P are 8 more than girls in Q.

Girls in R are 40% more than girls in Q.

Boys in R are $\frac{3}{8}$ of the girls in R.

Boys in Q are 14 more than girls in P.

Q7. Find the difference between boys in P and Q together and girls in Q and R together.

- (a) 8
- (b) 12
- (c) 16
- (d) 10
- (e) 14

Answer: (b) 12

Solution:

$$\text{Boys in P} + \text{Q} = 22 + 62 = 84$$

$$\text{Girls in Q} + \text{R} = 40 + 56 = 96$$

$$\text{Difference} = 96 - 84 = 12$$

Q8. Boys in college P are what percentage of the total students in college R?

- (a) 25%
- (b) 30.77%
- (c) 28.57%
- (d) 33.33%

(e) 22.5%

Answer: (c) 28.57%

Solution:

Boys in P = 55% of 40 = 22

Girls in R = $1.4 \times 40 = 56$; Boys in R = $\frac{3}{8} \times 56 = 21$

Total students in R = $21 + 56 = 77$

Required percentage = $\frac{22}{77} \times 100 = 28.57\%$ approximately

Q9. Find the ratio of girls in P to boys in R.

(a) 12:5

(b) 16:7

(c) 18:7

(d) 15:8

(e) 20:9

Answer: (b) 16:7

Solution:

Girls in P = $40 + 8 = 48$

Boys in R = 21

Required ratio = $48 : 21 = 16 : 7$

Q10. The total number of students in college W is 30% more than the total number of students in college P. The ratio of boys in R to boys in W is 3:7. Find the number of girls in W.

(a) 38

(b) 40

(c) 44

(d) 42

(e) 46

Answer: (d) 42

Solution:

Students in P = $22 + 48 = 70$

Students in W = $130\% \times 70 = 91$

Boys in R = 21; Boys R : Boys W = 3:7

Boys in W = $21 \times \frac{7}{3} = 49$

Girls in W = $91 - 49 = 42$

Q11.

Q11. The ratio of milk to water in Jar A is 5:2, while the ratio in Jar B is 3:8. The quantity of water in Jar B is 20 litres more than the quantity of water in Jar A. After mixing both jars, the total quantity of milk is 19 litres more than the total quantity of water. Find the quantity of milk in the final mixture.

(a) 101 litres

(b) 121 litres

(c) 111 litres

(d) 131 litres

(e) 91 litres

Answer: (c) 111 litres

Solution:

Let Jar A contain: Milk = $5x$, Water = $2x$

Jar B: Milk = $3y$, Water = $8y$

Given: Water B – Water A = 20 $\rightarrow 8y - 2x = 20 \rightarrow 4y - x = 10 \dots(1)$

Also, Total milk – Total water = 19

$(5x + 3y) - (2x + 8y) = 19 \rightarrow 3x - 5y = 19 \dots(2)$

From (1): $x = 4y - 10$

Substitute in (2): $3(4y - 10) - 5y = 19 \rightarrow 12y - 30 - 5y = 19 \rightarrow 7y = 49 \rightarrow y = 7, x = 18$

Milk in Jar A = $5 \times 18 = 90$ litres; Milk in Jar B = $3 \times 7 = 21$ litres

Total milk = $90 + 21 = 111$ litres

Directions (Q12–Q14): Solve both equations and establish the relationship between x and y .

(a) $x > y$ (b) $x \geq y$ (c) $x < y$ (d) $x \leq y$ (e) $x = y$ or no relation can be established

Q12. I. $2x^2 - 7x + 3 = 0$

II. $y^2 - 7y + 10 = 0$

(a) $x > y$

(b) $x \geq y$

(c) $x < y$

(d) $x \leq y$

(e) $x = y$ or no relation can be established

Answer: (e) $x = y$ or no relation can be established

Solution:

I. $2x^2 - 7x + 3 = 0 \rightarrow (2x - 1)(x - 3) = 0 \rightarrow x = 1/2, 3$

II. $y^2 - 7y + 10 = 0 \rightarrow (y - 2)(y - 5) = 0 \rightarrow y = 2, 5$

Since some values of x are greater than y while others are smaller, no definite relationship can be established.

Q13. I. $x^2 - 9x + 20 = 0$

II. $y^2 - 3y + 2 = 0$

(a) $x > y$

(b) $x \geq y$

(c) $x < y$

(d) $x \leq y$

(e) $x = y$ or no relation can be established

Answer: (a) $x > y$

Solution:

I. $(x - 4)(x - 5) = 0 \rightarrow x = 4, 5$

II. $(y - 1)(y - 2) = 0 \rightarrow y = 1, 2$

Every possible value of x is greater than every possible value of y .

Therefore, $x > y$.

Q14. I. $x^2 - 5x + 6 = 0$

II. $y^2 - 9y + 18 = 0$

- (a) $x > y$
- (b) $x \geq y$
- (c) $x < y$
- (d) $x \leq y$
- (e) $x = y$ or no relation can be established

Answer: (d) $x \leq y$

Solution:

I. $(x - 2)(x - 3) = 0 \rightarrow x = 2, 3$

II. $(y - 3)(y - 6) = 0 \rightarrow y = 3, 6$

Every possible value of x is less than or equal to every possible value of y .

Therefore, $x \leq y$.

Directions (Q15–Q18): Find the approximate value of the question mark (?).

Q15. 34.89% of $200.15 + 19.92\%$ of $349.87 - 105.20 = ?$

- (a) 25
- (b) 45
- (c) 30
- (d) 35
- (e) 40

Answer: (d) 35

Solution:

Approximate: 35% of $200 + 20\%$ of $350 - 105$

$= 70 + 70 - 105$

$= 35$

Q16. $63.90 \div 7.98 \times 14.95 + 221.85 = ?^2 + 13.10 \times 8.95$

- (a) 12
- (b) 18
- (c) 15
- (d) 20
- (e) 10

Answer: (c) 15

Solution:

Approximate: $64 \div 8 \times 15 + 222 = 8 \times 15 + 222 = 120 + 222 = 342$

And, $13 \times 9 = 117$

Therefore, $?^2 \approx 342 - 117 = 225$

$? \approx \sqrt{225} = 15$

Q17. ?% of $(949.80 - 249.70 - 179.90 - 119.60) = 99.60$

- (a) 20
- (b) 30
- (c) 25
- (d) 35
- (e) 40

Answer: (c) 25

Solution:

Approximate expression: $950 - 250 - 180 - 120 = 400$

So, ?% of 400 = 100

$\frac{?}{100} \times 400 = 100$

? = 25

Q18. $(? - 380.20) - 45.1\%$ of $(1499.70 \div 14.98) = 16.10 \times 21.90 \div 87.80$

- (a) 419
- (b) 439
- (c) 449
- (d) 429
- (e) 409

Answer: (d) 429

Solution:

Approximate: $1500 \div 15 = 100$

45% of 100 = 45

Right side: $16 \times 22 \div 88 = 4$

Therefore, $? - 380 - 45 = 4$

? = 429

Directions (Q19–Q24): The following table shows the average number of students visiting Schools M and N on five different days.

Additional information:

On Monday, students visiting from M are 12 more than those from N.

The ratio of students visiting from N from Monday to Friday is 5:3:4:2:6.

Since average = $(M+N)/2$:

Day	Average of M & N
Monday	66
Tuesday	60
Wednesday	78
Thursday	51
Friday	69

Day	M+N	M	N
Monday	132	72	60
Tuesday	120	84	36
Wednesday	156	108	48

Thursday	102	78	24
Friday	138	66	72

Q19. On Monday, the ratio of boys to girls visiting from M is 7:5. The number of boys visiting from N on Friday is equal to the number of girls visiting from M on Monday. Find the girls visiting from N on Friday as a percentage of the students visiting from M on Tuesday.

- (a) 40%
- (b) 45%
- (c) 50%
- (d) 55%
- (e) 60%

Answer: (c) 50%

Solution:

M students on Monday = 72

Boys : Girls = 7:5 $\rightarrow$ Girls from M = $72 \times \frac{5}{12} = 30$

Therefore, boys from N on Friday = 30

Total students from N on Friday = 72

Girls from N on Friday = $72 - 30 = 42$

Required percentage = $\frac{42}{84} \times 100 = 50\%$

Q20. The total number of students in School O is 30% more than the average number of students visiting from N on Monday, Wednesday and Friday. The ratio of total students in O to P is 6:7. Find the number of students in P.

- (a) 84
- (b) 98
- (c) 91
- (d) 105
- (e) 77

Answer: (c) 91

Solution:

Average N students = $(60 + 48 + 72)/3 = 180/3 = 60$

Students in O = $130\% \times 60 = 78$

O : P = 6 : 7

P = $78 \times \frac{7}{6} = 91$

Q21. Find the average number of students visiting from School N on all five days.

- (a) 44
- (b) 46
- (c) 50
- (d) 52
- (e) 48

Answer: (e) 48

Solution:

$$\text{Total N students} = 60 + 36 + 48 + 24 + 72 = 240$$

$$\text{Average} = 240/5 = 48$$

Q22. Students from M visiting on Wednesday participated in two games — football and hockey — in the ratio 2:1. Find the difference between students participating in hockey and the total number of students visiting from both schools on Thursday.

- (a) 60
- (b) 70
- (c) 64
- (d) 62
- (e) 66

Answer: (e) 66

Solution:

$$\text{M students on Wednesday} = 108$$

$$\text{Football : Hockey} = 2 : 1 \rightarrow \text{Hockey players} = 108 \times 1/3 = 36$$

$$\text{Total M+N students on Thursday} = 102$$

$$\text{Required difference} = 102 - 36 = 66$$

Q23. The number of students visiting from M on Tuesday is what percentage more than the students visiting from N on Friday?

- (a) 12.5%
- (b) 20%
- (c) 18.5%
- (d) 14.5%
- (e) 16.67%

Answer: (e) 16.67%

Solution:

$$\text{M on Tuesday} = 84$$

$$\text{N on Friday} = 72$$

$$\text{Difference} = 12$$

$$\text{Required percentage} = 12/72 \times 100 = 16.67\%$$

Q24. On a particular week, 1/3 of the students scheduled to visit from M on Monday did not visit, while 25% of those scheduled from N on Wednesday did not visit. Find the total number of students who did not visit.

- (a) 30
- (b) 32
- (c) 34
- (d) 36
- (e) 38

Answer: (d) 36

Solution:

$$\text{M on Monday} = 72$$

Non-visitors = $\frac{1}{3}$ of 72 = 24
N on Wednesday = 48
Non-visitors = 25% of 48 = 12
Total non-visitors = 24 + 12 = 36

Directions (Q25–Q30): Study the following data carefully and answer the questions.

The table shows the total population (males + females) of four cities E, F, G and H in 2024 and 2025.

City	2024	2025
E	600	1000
F	550	900
G	640	760
H	858	440

Q25. The ratio of males to females in city E in 2024 is 5:7. The number of females in city E in 2025 is 1.6 times the number of females in 2024. Find the difference between the number of males in city E in 2024 and 2025.

- (a) 170
- (b) 200
- (c) 190
- (d) 210
- (e) 180

Answer: (c) 190

Solution:

Males in E in 2024 = $600 \times \frac{5}{12} = 250$
Females in 2024 = $600 - 250 = 350$
Females in 2025 = $350 \times 1.6 = 560$
Males in 2025 = $1000 - 560 = 440$
Required difference = $440 - 250 = 190$

Q26. The population of city J in 2024 is 20% more than that of city F in 2024. If the total population of city J in 2024 and 2025 together is 1500, find its population in 2025.

- (a) 800
- (b) 820
- (c) 860
- (d) 840
- (e) 880

Answer: (d) 840

Solution:

Population of J in 2024 = $550 \times \frac{120}{100} = 660$
Population of J in 2025 = $1500 - 660 = 840$

Q27. The population of city F in 2023 was 20% less than the average population of cities E and G in 2025. Find the ratio of the population of F in 2023 to that of H in 2024.

- (a) 29:39
- (b) 30:41
- (c) 31:38
- (d) 33:40
- (e) 32:39

Answer: (e) 32:39

Solution:

Average population of E and G in 2025 = $(1000 + 760)/2 = 880$

F's population in 2023 = 80% of 880 = 704

Required ratio = 704 : 858 = 32 : 39

Q28. The number of males in city H in 2024 is 40% of the total population of city F in 2025. The number of males in H in 2025 is two-thirds the number of males in H in 2024. Find the total number of females in H in 2024 and 2025 together.

- (a) 678
- (b) 718
- (c) 698
- (d) 708
- (e) 688

Answer: (c) 698

Solution:

Males in H in 2024 = 40% of 900 = 360

Females in H in 2024 = $858 - 360 = 498$

Males in H in 2025 = $360 \times 2/3 = 240$

Females in H in 2025 = $440 - 240 = 200$

Required total = $498 + 200 = 698$

Q29. The population of city G in 2021 was 60% of the population of city E in 2024. If 45% of the population of G in 2021 were males, find the number of females as a percentage of the population of city H in 2025.

- (a) 40%
- (b) 48%
- (c) 42%
- (d) 45%
- (e) 36%

Answer: (d) 45%

Solution:

Population of G in 2021 = 60% of 600 = 360

Males = 45% of 360 = 162

Females = $360 - 162 = 198$

Required percentage = $198/440 \times 100 = 45\%$

Q30. $\frac{5}{6}$ of the population of city H in 2024 are literate, while 30% of the population of city F in 2024 are illiterate. Find the difference between the total literate population of F and H together and the total illiterate population of F and H together.

- (a) 772
- (b) 782
- (c) 802
- (d) 812
- (e) 792

Answer: (e) 792

Solution:

$$\text{Literate in H} = 858 \times \frac{5}{6} = 715$$

$$\text{Illiterate in H} = 858 - 715 = 143$$

$$\text{Illiterate in F} = 30\% \text{ of } 550 = 165$$

$$\text{Literate in F} = 550 - 165 = 385$$

$$\text{Total literates} = 715 + 385 = 1100$$

$$\text{Total illiterates} = 143 + 165 = 308$$

$$\text{Difference} = 1100 - 308 = 792$$

Directions (Q31–Q34): Read the following information carefully.

There are 458 students in three colleges S, T and U.

Number of girls in T = 60.

Boys in S are 80% of girls in T.

Girls in S are 30 more than girls in T.

Girls in U are 1.5 times the girls in T.

Boys in U are $\frac{5}{9}$ of the girls in U.

Boys in T are 30 more than girls in S.

Q31. Find the difference between boys in S and T together and girls in T and U together.

- (a) 12
- (b) 14
- (c) 16
- (d) 20
- (e) 18

Answer: (e) 18

Solution:

$$\text{Boys in S} + \text{T} = 48 + 120 = 168$$

$$\text{Girls in T} + \text{U} = 60 + 90 = 150$$

$$\text{Difference} = 168 - 150 = 18$$

Q32. Boys in college S are what percentage of the total students in college U?

- (a) 30%
- (b) 37.5%

- (c) 32%
- (d) 34.29%
- (e) 36%

Answer: (d) 34.29%

Solution:

Boys in S = 80% of 60 = 48

Girls in U = $1.5 \times 60 = 90$; Boys in U = $5/9 \times 90 = 50$

Total students in U = $50 + 90 = 140$

Required percentage = $48/140 \times 100 = 34.29\%$ approximately

Q33. Find the ratio of girls in S to boys in U.

- (a) 7:4
- (b) 2:1
- (c) 11:6
- (d) 5:3
- (e) 9:5

Answer: (e) 9:5

Solution:

Girls in S = $60 + 30 = 90$

Boys in U = 50

Required ratio = $90 : 50 = 9 : 5$

Q34. The total number of students in college V is 50% more than the total number of students in college S. The ratio of boys in U to boys in V is 5:9. Find the number of girls in V.

- (a) 109
- (b) 113
- (c) 121
- (d) 125
- (e) 117

Answer: (e) 117

Solution:

Students in S = $48 + 90 = 138$

Students in V = $150\% \times 138 = 207$

Boys in U = 50; Boys U : Boys V = 5:9

Boys in V = $50 \times 9/5 = 90$

Girls in V = $207 - 90 = 117$

Q35.

Q35. The ratio of milk to water in Jar C is 7:3, while the ratio in Jar D is 2:5. The quantity of water in Jar D is 22 litres more than the quantity of water in Jar C. After mixing both jars, the total quantity of milk is 11 litres more than the total quantity of water. Find the quantity of water in the final mixture.

- (a) 78 litres

- (b) 98 litres
- (c) 88 litres
- (d) 108 litres
- (e) 68 litres

Answer: (c) 88 litres

Solution:

Let Jar C contain: Milk = $7x$, Water = $3x$

Jar D: Milk = $2y$, Water = $5y$

Given: Water D – Water C = 22 $\rightarrow 5y - 3x = 22 \dots(1)$

Also, Total milk – Total water = 11

$(7x + 2y) - (3x + 5y) = 11 \rightarrow 4x - 3y = 11 \dots(2)$

Solving (1) and (2): $x = 11$, $y = 11$

Water in Jar C = $3 \times 11 = 33$ litres; Water in Jar D = $5 \times 11 = 55$ litres

Total water = $33 + 55 = 88$ litres

Directions (Q36–Q38): Solve both equations and establish the relationship between x and y .

(a) $x > y$ (b) $x \geq y$ (c) $x < y$ (d) $x \leq y$ (e) $x = y$ or no relation can be established

Q36. I. $2x^2 - 11x + 12 = 0$

II. $y^2 - 7y + 6 = 0$

- (a) $x > y$
- (b) $x \geq y$
- (c) $x < y$
- (d) $x \leq y$
- (e) $x = y$ or no relation can be established

Answer: (e) $x = y$ or no relation can be established

Solution:

I. $2x^2 - 11x + 12 = 0 \rightarrow (2x - 3)(x - 4) = 0 \rightarrow x = 3/2, 4$

II. $y^2 - 7y + 6 = 0 \rightarrow (y - 1)(y - 6) = 0 \rightarrow y = 1, 6$

Since some values of x are greater than y while others are smaller, no definite relationship can be established.

Q37. I. $x^2 - 3x + 2 = 0$

II. $y^2 - 12y + 35 = 0$

- (a) $x > y$
- (b) $x \geq y$
- (c) $x < y$
- (d) $x \leq y$
- (e) $x = y$ or no relation can be established

Answer: (c) $x < y$

Solution:

I. $(x - 1)(x - 2) = 0 \rightarrow x = 1, 2$

II. $(y - 5)(y - 7) = 0 \rightarrow y = 5, 7$

Every possible value of x is less than every possible value of y .
Therefore, $x < y$.

Q38. I. $x^2 - 10x + 24 = 0$

II. $y^2 - 6y + 8 = 0$

- (a) $x > y$
- (b) $x \geq y$
- (c) $x < y$
- (d) $x \leq y$
- (e) $x = y$ or no relation can be established

Answer: (b) $x \geq y$

Solution:

I. $(x - 4)(x - 6) = 0 \rightarrow x = 4, 6$

II. $(y - 2)(y - 4) = 0 \rightarrow y = 2, 4$

Every possible value of x is greater than or equal to every possible value of y .
Therefore, $x \geq y$.

Directions (Q39–Q42): Find the approximate value of the question mark (?).

Q39. $44.90\% \text{ of } 279.80 + 15.10\% \text{ of } 419.70 - 87.80 = ?$

- (a) 91
- (b) 96
- (c) 101
- (d) 106
- (e) 111

Answer: (c) 101

Solution:

$$\begin{aligned} &\text{Approximate: } 45\% \text{ of } 280 + 15\% \text{ of } 420 - 88 \\ &= 126 + 63 - 88 \\ &= 101 \end{aligned}$$

Q40. $71.85 \div 7.95 \times 23.90 + 50.90 = ?^2 + 14.10 \times 6.95$

- (a) 11
- (b) 17
- (c) 15
- (d) 19
- (e) 13

Answer: (e) 13

Solution:

$$\begin{aligned} &\text{Approximate: } 72 \div 8 \times 24 + 51 = 9 \times 24 + 51 = 216 + 51 = 267 \\ &\text{And, } 14 \times 7 = 98 \\ &\text{Therefore, } ?^2 \approx 267 - 98 = 169 \\ &? \approx \sqrt{169} = 13 \end{aligned}$$

Q41. ?% of $(650.20 - 150.40 - 99.80 - 49.90) = 104.80$

- (a) 20
- (b) 35
- (c) 25
- (d) 30
- (e) 40

Answer: (d) 30

Solution:

Approximate expression: $650 - 150 - 100 - 50 = 350$

So, ?% of 350 = 105

$\frac{?}{100} \times 350 = 105$

? = 30

Q42. $(? - 299.80) - 35.2\% \text{ of } (1799.60 \div 17.95) = 19.90 \times 18.10 \div 39.85$

- (a) 334
- (b) 354
- (c) 364
- (d) 324
- (e) 344

Answer: (e) 344

Solution:

Approximate: $1800 \div 18 = 100$

35% of 100 = 35

Right side: $20 \times 18 \div 40 = 9$

Therefore, $? - 300 - 35 = 9$

? = 344

Directions (Q43–Q48): The following table shows the average number of students visiting Schools R and S on five different days.

Additional information:

On Monday, students visiting from R are 56 more than those from S.

The ratio of students visiting from S from Monday to Friday is 7:5:6:4:8.

Since average = $(R+S)/2$:

Day	Average of R & S
Monday	84
Tuesday	56
Wednesday	54
Thursday	54
Friday	66

Day	R+S	R	S
Monday	168	112	56

Tuesday	112	72	40
Wednesday	108	60	48
Thursday	108	76	32
Friday	132	68	64

Q43. On Monday, the ratio of boys to girls visiting from R is 9:7. The number of boys visiting from S on Friday is equal to the number of girls visiting from R on Monday. Find the girls visiting from S on Friday as a percentage of the students visiting from R on Wednesday.

- (a) 20%
- (b) 30%
- (c) 25%
- (d) 35%
- (e) 22.5%

Answer: (c) 25%

Solution:

R students on Monday = 112

Boys : Girls = 9:7 $\rightarrow$ Girls from R = $112 \times 7/16 = 49$

Therefore, boys from S on Friday = 49

Total students from S on Friday = 64

Girls from S on Friday = $64 - 49 = 15$

Required percentage = $15/60 \times 100 = 25\%$

Q44. The total number of students in School T is 25% more than the average number of students visiting from S on Monday, Wednesday and Friday. The ratio of total students in T to U is 5:7. Find the number of students in U.

- (a) 84
- (b) 91
- (c) 105
- (d) 98
- (e) 112

Answer: (d) 98

Solution:

Average S students = $(56 + 48 + 64)/3 = 168/3 = 56$

Students in T = $125\% \times 56 = 70$

T : U = 5 : 7

U = $70 \times 7/5 = 98$

Q45. Find the average number of students visiting from School S on all five days.

- (a) 44
- (b) 46
- (c) 50
- (d) 52
- (e) 48

Answer: (e) 48

Solution:

$$\text{Total S students} = 56 + 40 + 48 + 32 + 64 = 240$$

$$\text{Average} = 240/5 = 48$$

Q46. Students from R visiting on Wednesday participated in two games — football and hockey — in the ratio 3:1. Find the difference between students participating in hockey and the total number of students visiting from both schools on Thursday.

- (a) 83
- (b) 103
- (c) 98
- (d) 88
- (e) 93

Answer: (e) 93

Solution:

$$\text{R students on Wednesday} = 60$$

$$\text{Football : Hockey} = 3:1 \rightarrow \text{Hockey players} = 60 \times 1/4 = 15$$

$$\text{Total R+S students on Thursday} = 108$$

$$\text{Required difference} = 108 - 15 = 93$$

Q47. The number of students visiting from R on Tuesday is what percentage more than the students visiting from S on Friday?

- (a) 20%
- (b) 12.5%
- (c) 15%
- (d) 10%
- (e) 18.75%

Answer: (b) 12.5%

Solution:

$$\text{R on Tuesday} = 72$$

$$\text{S on Friday} = 64$$

$$\text{Difference} = 8$$

$$\text{Required percentage} = 8/64 \times 100 = 12.5\%$$

Q48. On a particular week, 25% of the students scheduled to visit from R on Monday did not visit, while 1/8 of those scheduled from S on Wednesday did not visit. Find the total number of students who did not visit.

- (a) 28
- (b) 30
- (c) 32
- (d) 34
- (e) 36

Answer: (d) 34

Solution:

R on Monday = 112

Non-visitors = 25% of 112 = 28

S on Wednesday = 48

Non-visitors = $\frac{1}{8}$ of 48 = 6

Total non-visitors = $28 + 6 = 34$